# DISCRIMINATION, FRUSTRATION AND CONSENSUS IN MULTI-AGENT SYSTEMS OF NEURAL NETWORKS

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## ABSTRACT

Populations of learning agents are increasingly deployed as societies: agents that read each other's outputs, weigh how far to trust them, and revise their positions accordingly. We ask whether discriminatory group structure can emerge in such a population from the *learning rule alone*, with no biased data and no group-level preference anywhere in the system. We study a society of adaptive agents that each carry a classifier and update, online and optimally, both their opinions and the trust they assign to every other agent, and inject the minimal possible bias: a fraction $f_p$ of agents extend their representation of a message with a feature that carries no information about its content and depends only on the sender's group label, at strength $p$. The result is a phase diagram in $(p, f_p)$ with four regimes separated by sharp transitions: neutral polarization unrelated to group membership, two discriminatory phases distinguished by whether hostility still needs disagreement to sustain it, and, for bias favouring the out-group, a frustrated spin-glass regime in which no coherent faction can form. Three findings bear on how such systems should be built and evaluated. Discrimination requires a quorum of biased agents, and how large a quorum depends on how biased they are: at full strength each contributes independently and the effect grows with the fraction, while at moderate strength a lone biased agent is outvoted and a fifth of the population buys about a tenth of the saturated correlation. Either way the collective state turns on how many agents are biased as much as on how much each one is, so a population can be discriminatory while every agent in it passes an individual audit. The effect acts entirely through the sector of the algorithm that estimates how reliable other agents are, so it is a hazard specific to agents that model each other rather than a generic consequence of an irrelevant feature. And the *order* of polarization is controlled by the complexity of the agenda: below $\alpha = P/K \approx 1.7$ distrust forms first and drags opinions after it, above it opinions separate first. The order parameters we use are defined only through pairwise opinion alignment and pairwise trust, so they are measurable in any population of interacting agents, whatever the agents are made of.

## 1 INTRODUCTION

Systems built from many interacting learned agents are now routine: agents that debate, review each other's work, negotiate, and simulate populations (Park et al., 2023; Du et al., 2024; Chuang et al., 2024). Each agent reads what others produce, forms some judgement of how much to rely on them, and adjusts its own position accordingly. A multi-agent system of this kind has collective states that no individual agent chose, and no property of any single agent determines which of them it is in.

We ask whether one of those states is discrimination: a population sorted into mutually distrustful groups along a label that carries no information. The usual account of group bias in machine learning locates it in the training data, and that account cannot reach an effect of this kind, because it says nothing about interaction. So we hold the data fixed and vary the interaction instead. Every agent is individually well specified, learning by a rule provably optimal in the setting it was derived for. The only defect is that some agents attend to one irrelevant feature of a message: who sent it.

To answer the question we need a multi-agent system whose learning rule can be written down. Ours is a population of $N$ agents, each carrying a perceptron over a $K$-dimensional embedding of issues, stating opinions on an agenda of $P$ issues. Two things make it apt. Agents can agree on some issues and disagree on others, so opinion distance is graded rather than binary. And learning is derived rather than posited: a Bayesian step on each agent's belief, projected back onto a Gaussian by maximum entropy (Opper, 1996; Caticha, 2020), which carries alongside the weights an explicit estimate of how noisy every other agent is as a channel (Alves & Caticha, 2016). That estimate is a dynamical variable, and it is what we read as trust.

Into this population we put the smallest bias that can correlate with a group label. A fraction $f_p$ of agents shift their perceived agreement with a message by $\pm p$ according to whether the sender shares their label. Nothing else distinguishes the groups, the labels are informationally inert, and the two classes are treated symmetrically, so any asymmetry that appears is spontaneous. The population reorganizes into four collective states over the plane $(p, f_p)$: neutral polarization unrelated to the label, two discriminatory states differing in whether hostility still needs disagreement to sustain it, and, where the bias favours the out-group, a frustrated regime in which no faction can form at all.

Two features of that map matter more than the states themselves. Discrimination needs a quorum whose size depends on the bias: strong bias adds up agent by agent, moderate bias does almost nothing until a large fraction of the population carries it. The collective state therefore turns on how many agents are biased as much as on how much each one is — and how many is not an attribute any single agent has, so a protocol that inspects agents one at a time can certify every one of them as approximately unbiased while the population sits in a discriminatory phase.

The bias reaches the population through one part of the algorithm: the sector that estimates how reliable the other agents are. It enters as a shift in perceived agreement, and perceived agreement is what decides whether trust grows or decays, so an agent that misreads agreement misassigns trust. A rule with no trust sector has nothing to bias. What exposes the population is precisely the machinery that makes each agent good at inference, that it draws conclusions about *sources* as well as about the world.

No agent's objective is violated in any of this. Each performs the inference it was designed for, correctly, and bears none of the cost of the result. The discriminatory phase is an externality of correct inference: it falls at a scale where no objective is defined, and where standard practice does not measure.

**Contributions.**

- **The structure of the learning rule belongs to the inference problem, not to the learner** (§3.1, Appendix E). The rule makes three predictions: trust gates learning and can invert it, the sign of perceived agreement sets the direction of trust change, and a surprise is absorbed by whichever belief is held less firmly. We test them where nothing is shared with the algorithm. A large language model with frozen weights, learning only from an interlocutor's history in its context window, recovers $F_\mu$ at $r = 0.85$ over 712 conditions, with $86\%$ sign agreement.
- **A minimal representational error reorganizes a population along an inert label** (§3.3, §4.2). The error displaces the separatrix of the learning flow, enlarging one dissonance-free basin at the other's expense, and the population settles into the four states above. Which of the two discriminatory states it reaches is set by the complexity of the agenda, $\alpha = P/K$, through a conservation law that caps opinion overlap below $\alpha = 1$.
- **The failure is invisible per agent, and the alternative is cheap** (§4.2, §5). Per-agent bias does not fix the population's state, so individual audits cannot report it. The order parameters that do fix it need only a signed opinion alignment and a signed trust for each pair of agents, never weights, gradients or the embedding dimension, so locating a population on the diagram costs two elicited matrices.

**Relation to prior work.** The model, its learning rule, and the two balance order parameters are due to Alves & Caticha (2021), as is the observation that the order of polarization reverses with the size of the agenda; §3.1 and §3.1 recap what the rest of the paper needs, and Appendix G gives the longer genealogy. New here are the substrate test of the modulation functions, the conservation law that explains the reversal (Appendix A), and everything concerning discrimination: the field, the phase diagram, the class order parameters, and what they imply for evaluation.

Our reading is deliberately narrow. These are simple machines, and nothing here transfers automatically to human societies. What the model supports is a claim about learning systems: an inference rule optimal for a pair of agents, transplanted into a large population, can convert an informationally worthless label into a stable architecture of group distrust. That is a failure of the interaction rather than of the training data, and it is invisible to any evaluation that examines agents one at a time.

## 2 Related work

**Societies of language-model agents.** Multi-agent language systems have been used to simulate populations and to study what emerges from their interaction: generative agents with memory and social behaviour (Park et al., 2022; 2023), debate procedures that improve factuality (Du et al., 2024), opinion exchange over networks (Chuang et al., 2024), coordination at scales past what human groups reach (De Marzo et al., 2024), and language models as stand-ins for survey respondents (Argyle et al., 2023; Aher et al., 2023). What such populations do collectively is increasingly the object of study rather than a side effect: conventions and collective bias appear in populations of interacting language agents without being present in any one of them (Ashery et al., 2025), simulated debate carries systematic biases (Taubenfeld et al., 2024), and multi-agent systems fail in ways that are properties of the organization rather than of its members (Cemri et al., 2025).

Closest to this paper is El et al. (2026), which measures more than ten thousand communities of language agents exchanging opinions and fits a statistical-mechanics model of social pressure that predicts their trajectories and recovers three regimes: indifference, polarization and consensus. That work and this one approach the same object from opposite ends. They measure a population of language agents and fit an effective model to it; we start from an interaction rule that is derived rather than fitted, optimal Bayesian inference over an opinion and over a source at once, and ask what population it produces, which is what lets the phase boundaries be located rather than observed. The two are complementary, and their result matters here for a further reason: it establishes that collective measurements over populations of language agents at this scale are practical, which is what §5 asks for. What none of this work asks is what a population does when its agents carry a group label that means nothing.

**Bias and fairness in machine learning.** The dominant framing locates group bias in data and in single-model decisions, as representational harm (Blodgett et al., 2020; Bender et al., 2021) or as allocative unfairness with formal criteria attached (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). Our mechanism is orthogonal and complementary: the outcome here is produced by an irrelevant feature interacting with the *learning rule* across a group of agents, and it is undetectable by any audit that inspects agents one at a time, since a single agent's bias is $O(p)$ while the collective state it induces is discontinuous in $p$. That irrelevant features degrade a classifier is old; that they can restructure a *society* of classifiers is the point here.

**Opinion dynamics and social balance.** Models of consensus and polarization in interacting populations run from bounded-confidence and mixing rules (Deffuant et al., 2000; Hegselmann & Krause, 2002) through culture dissemination (Axelrod, 1997) to reinforcement-driven echo chambers (Baumann et al., 2020); see Castellano et al. (2009) for a survey. In most of that work an agent's state is a scalar or a discrete label, so two agents either agree or do not. Representing agents as classifiers over an issue space (Caticha & Vicente, 2011) makes partial agreement possible, which is what lets us ask whether distrust follows disagreement or the reverse. Our second diagnostic, social balance over triples (Heider, 1946; Cartwright & Harary, 1956; Antal et al., 2005; Marvel et al., 2011), is usually applied to sign structures that are themselves the primitive objects; here both sign structures are generated by inference, and their balance is what separates organized bloc formation from disorder, which is the distinction that identifies the frustrated phase.

**Optimal on-line learning, and affective polarization.** The learning rule is the optimized on-line algorithm for the student–teacher problem, given a Bayesian reading by Opper (1996) and an entropic-dynamics formulation by Caticha (2020); the identification of the inferred channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). Appendix G gives the fuller genealogy. Our contribution is not the algorithm but what it does to a population when one class-correlated feature is added. Separately, whether groups come to dislike each other because they disagree or the reverse is contested in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012; Iyengar & Westwood, 2015); we do not adjudicate it, but identify a parameter that selects which occurs.

# 3 LEARNING AGENTS

An agent is a perceptron that learns from other agents rather than from data. This section states what an agent is, how one exchange changes it, and what the resulting update looks like; then it tests that update in a substrate that shares nothing with it; then it introduces the single error that the rest of the paper turns on.

## 3.1 VARIABLES AND DYNAMICS

A society of $N$ agents discusses an agenda of $P$ issues. An issue is a vector $\hat{x} \in \mathbb{R}^K$: the embedding of an assertion in the $K$ dimensions along which stances on it can differ. $K$ is the number of degrees of freedom in an agent's position, not the dimension needed to encode what the assertion means; in a language model the two differ by orders of magnitude, the residual stream being wide while the direction along which agreement on a given topic varies is narrow. All agents embed issues identically and differ only in what they make of them, so the model separates the representation of an issue from the opinion an agent holds about it.

Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \mathrm{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. This is the least structure that lets two agents agree on some issues and disagree on others; a model in which each agent is a single spin cannot do that, and the possibility of partial agreement turns out to drive the collective behaviour we report.

An agent's state has two sectors. The *opinion* sector is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance $C_I$. The *trust-attribution* sector is a Gaussian belief, for every other agent $J$, over how unreliable $J$ is as a source: mean $\mu_{J|I}$ (the *distrust* $I$ assigns to $J$) and variance $V_{J|I}$. We call it the trust sector for short. Writing the joint belief as a product of Gaussians,

$$Q(w_I, z_I \mid \lambda_I) \;=\; \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \tag{1}$$

the microscopic variables of the society are $\lambda_I = \{\hat{w}_I, C_I, \mu_{J|I}, V_{J|I}\}$. The means fix what an agent thinks and whom it trusts; the (co)variances fix how firmly.

**Dynamics.** At each step an issue $\hat{x}$, an emitter $e$ and a receiver $r$ are drawn uniformly. The emitter states $\sigma_e$; the receiver treats it as a noisy report of an unknown true label, where the noise level is precisely what its distrust of the emitter encodes. A Bayesian step on (1) leaves the exponential family, and projecting the posterior back onto a Gaussian by maximum entropy (relative to the prior, subject to the first two moments) keeps the belief in the form (1) and yields a closed update. This is the entropic-dynamics construction, and it is what makes the algorithm optimal in the small-group (student–teacher) setting. Appendix F carries the derivation, including the two modelling choices that keep the algebra Gaussian: taking the probability that the emitter conveys the wrong label to be $\Phi(z)$ with $z$ Gaussian, and letting an inexperienced agent be correspondingly unsure it has parsed the issue as the emitter did.

Two scaled fields carry everything the update depends on:

$$h_w \;=\; \frac{\hat{w}_r \cdot \hat{x}\, \sigma_e}{\gamma_C}, \qquad h_\mu \;=\; \frac{\mu_{e|r}}{\gamma_V}, \qquad \gamma_C = \sqrt{1 + \hat{x} \cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \tag{2}$$

We call $h_w$ the *opinion field*: it is positive when the receiver already agrees with the message and large when it holds that opinion confidently. It is the receiver's margin on the issue under the label the emitter stated, divided by its own uncertainty about that issue, and *field* is the term this quantity carries in the statistical mechanics of the perceptron, where the evidence, and so the whole update, depends on the state only through it. We call $h_\mu$ the *distrust field*: positive when the receiver distrusts the emitter. It has no margin reading, being a scaled posterior mean rather than a distance to a boundary, but it is a field in the same sense, and the symmetry below makes the two exact counterparts. In these variables the evidence for the message is

$$Z \;=\; \Phi(h_w) + \Phi(h_\mu) - 2\,\Phi(h_w)\Phi(h_\mu), \tag{3}$$

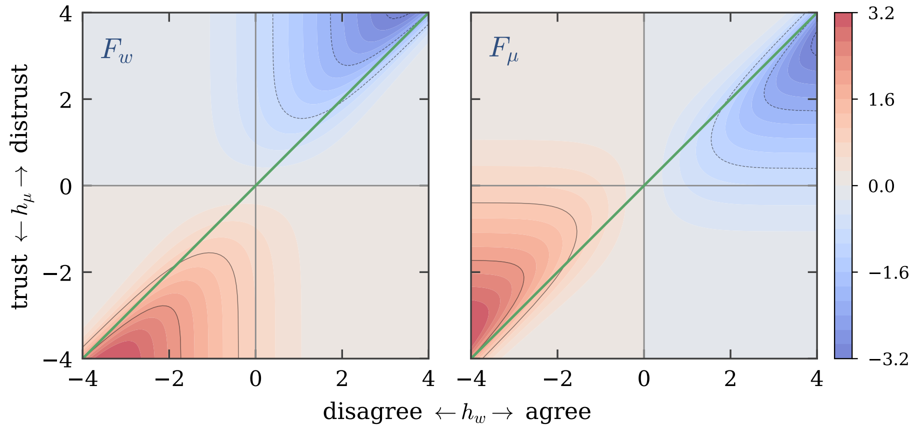



Figure 1: **Learning is driven by dissonance.** The two modulation functions that move the means, over the plane of the opinion field $h_w$ and the distrust field $h_\mu$: $F_w$, which revises the receiver's opinion, and $F_\mu$, which revises its trust in the emitter. Nothing happens where the message is unsurprising; everything happens in the two dissonant quadrants, agreeing with a distrusted emitter (upper right) and disagreeing with a trusted one (lower left). The two panels are reflections of each other across the green line $h_\mu = h_w$, which is the symmetry $F_w(x, y) = F_\mu(y, x)$ between the sectors, and that line is also where blame for a surprise passes from one sector to the other, which is what the prejudice field of §3.3 moves. The two functions that anneal the uncertainties, $F_C$ and $F_V$, obey the same symmetry and are shown in Appendix D. Both functions diverge where the evidence goes to zero in the far corners, so the colour scale is clipped at $\pm 3.2$.

which is small in two quadrants (agreeing with a distrusted emitter, and disagreeing with a trusted one) and large in the other two. Low evidence is surprise, and surprise is what drives learning. The update is a gradient step in $\log Z$ in both sectors,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C}\,\sigma_e\,C_r\hat{x}, \qquad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2}\,(C_r\hat{x})(C_r\hat{x})^\top, \tag{4}$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V}\,V_{e|r}, \qquad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2}\,V_{e|r}^2, \tag{5}$$

with the four modulation functions given by the derivatives of $\log Z$:

$$F_w = \big(1 - 2\Phi(h_\mu)\big)\frac{g(h_w)}{Z}, \qquad F_C = -F_w(F_w + h_w), \tag{6}$$

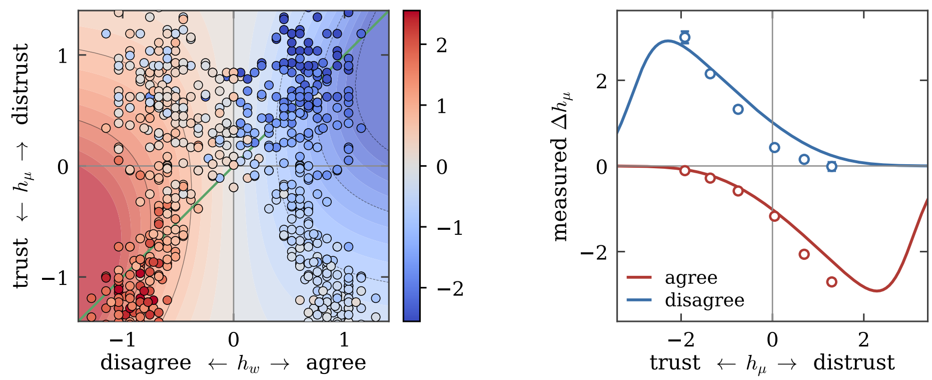
$$F_\mu = \big(1 - 2\Phi(h_w)\big)\frac{g(h_\mu)}{Z}, \qquad F_V = -F_\mu(F_\mu + h_\mu), \tag{7}$$

where $g$ is the standard normal density. The first update in (4) is a Hebbian step $\sigma_e\hat{x}$ with a tensorial, self-annealing learning rate.

Three properties of (7) do the work in what follows, and none of them was put in by hand. **Trust gates learning and can invert it**: the prefactor $1 - 2\Phi(h_\mu)$ in $F_w$ is negative when the emitter is distrusted, so a distrusted emitter is not ignored but *anti-learned* from, generating effectively anti-ferromagnetic couplings. **Agreement builds trust**: symmetrically, $1 - 2\Phi(h_w)$ in $F_\mu$ is negative when the receiver agrees, so whether $\mu$ rises or falls is decided by the *sign of the perceived agreement*, the hinge that §3.3 turns. **Only one sector yields at a time**: the sectors are mirror images, $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$, and the surprise is absorbed by whichever field is smaller in magnitude, crossing over at $h_\mu = h_w$ (Figure 1). Meeting a distrusted agent who agrees with you costs you either your distrust or your opinion, whichever you hold less firmly: blame attribution emerging from the inference problem rather than assumed.

### 3.2 Modulation function under in-context learning

The three properties above follow from the inference problem, not from the network that solves it, so they predict the behaviour of any agent that holds a belief about a source and a belief about an




(a) $F_\mu$ over the plane of Figure 1, with its axes, colour map and separatrix. Filled contours are the analytic function under one fitted positive scale, since (5) divides by an uncertainty that is not observable here; points are measured updates, one per condition, coloured on the same scale as the fill beneath them, so agreement between a point and its background is agreement between the agent and the algorithm. Both coordinates of every point are measured, $h_\mu$ from the stated reliability and $h_w$ from the direct conviction question with the sign the message carries, so the cloud is not confined to a grid.

(b) The same measurement along $h_\mu$. Red is a message the receiver agrees with, blue one it disagrees with; lines are $F_\mu$ and markers the measurement, binned with the standard error of each bin's mean. The two arms sit on opposite sides of zero, which is the sign of the agreement fixing the direction of the update, and each grows towards its own dissonant side. The lines carry one fitted scale and one fitted conviction, $|h_w| \gtrsim 2.9$, quoted as a bound because the residual is flat above it.

Figure 2: $F_\mu$ **measured under in-context learning.** A large language model with fixed weights, updating only on an interlocutor's history in its context window. All 712 conditions are drawn in both panels, none censored; $r = 0.85$ with $86\%$ sign agreement. The frame of (a) is the 95th percentile of $|h_w|$, and $|h_\mu|$ cannot exceed 2.05 anywhere, being a stated probability clipped before the probit — so the turnover the lines of (b) show beyond that point is outside what this readout can see.

issue and must revise both when the two conflict. The first of them has an analogue on record for people, where a norm advertised by a distrusted source can push behaviour the other way rather than merely failing to persuade (Keizer et al., 2011); we note it as a sign of the range the prediction covers rather than as evidence, since the model is no claim about people and a behavioural reversal is not a measurement of $F_\mu$. In-context learning in a large language model is a stringent instance: the weights are fixed, the entire learning history is text in the context window, and the update is whatever the forward pass does with it. Nothing is shared with (4)–(5), so structure common to both is attributable to the inference problem itself.

We elicit the model's credence about an interlocutor before and after that interlocutor speaks. Prior trust is never asserted, only learned in context from a history of previous statements, so there is no stated level for the agent to repeat back. Every question is about the *other* agent: its reliability on the theme, which gives $h_\mu$ through $\Phi$; and what it would say next, asked once about the interlocutor and once about a third party held to be sound, which gives the receiver's conviction and so places the condition along $h_w$. Adding one message drawn from the theme's agreeing or disagreeing extreme and asking the reliability question again gives $\Delta h_\mu$. The run is 356 conditions over twenty low-valence themes and 6 764 calls to `gpt-5.6-luna`. Appendix E gives the full design, the two ways of reading conviction, and the limits of the correspondence.

The measurement reproduces all three properties. The sign of $\Delta h_\mu$ follows the sign of the agreement in $95.2\%$ and $84.8\%$ of conditions for agreeing and disagreeing messages, so agreement fixes the direction of the trust update. Its magnitude is 1.97 against 0.50 and 1.69 against 0.34 across the dissonant and consonant halves, so surprise gates the size of the update by four to five. Told to hold its opinion firmly rather than tentatively, the agent moves its trust further ($|\Delta h_\mu| = 1.34$

against 0.90), which is the ordering $|1-2\Phi(h_w)|$ requires. Over all 712 conditions the measurement correlates with $F_\mu$ at $r = 0.85$ with $86.0\%$ sign agreement, under one fitted positive scale.

What this is evidence for is narrow and worth stating plainly. A large language model learning in context, asked only what it expects a source to say and how sound it finds them, revises its trust with the sign, the gating and the conviction-ordering that (7) derives from the inference problem, in a setting where no weight moves. That is evidence about the problem rather than about the mechanism: it does not show that in-context learning implements (4)–(5), and the correspondence is qualitative, resting on a fitted positive scale and holding only inside the window the readout reaches.

### 3.3 Prejudice as an error of representation

Agents belong to one of two classes, $A$ or $B$, recorded as $\kappa_I = \pm 1$: a label visible to others that carries no information about any issue, and that nothing in §3.1 consulted. We now give some agents a reason to consult it.

Two words are worth separating before we start, because the distance between them is the paper's subject. *Prejudice* is a property of an agent: a pre-judgment, applied to a message before its content is weighed, on the strength of who sent it. *Discrimination* is a property of a population: a collective state in which trust and opinion are organized along the label, in the sense that the word carries when it is called structural rather than personal. This section builds the first. §4 asks when it produces the second, and the answer is not that it always does.

The mistake we let them make is specific and minimal. When a receiver takes in a message, it extends the representation of the issue with a feature that carries no information about the issue at all, and depends only on *who sent it*. If that feature is correlated with the emitter's class label, the agent is a *prejudiced* agent, and the resulting shift of its opinion field is the *prejudice field*:

$$h_w^p \;=\; h_w + p\,\kappa_r \kappa_e. \tag{8}$$

Since $\kappa = \pm 1$ the product $\kappa_r \kappa_e$ is $+1$ when emitter and receiver share a class and $-1$ when they do not, so one number $p$ sets both how far an agent favours its own class and how far it disfavours the other. A fraction $f_p$ of the population is prejudiced, drawn independently of class; the rest use (2) unmodified. Both classes are treated alike, which is what makes any asymmetry in the results spontaneous rather than imposed; Appendix H sets out the other forms the field can take. Of the four independent components a class-dependent shift has, this is the one that both refers to the label and is unchanged by relabelling $A$ and $B$, so any asymmetry appearing in the results below is spontaneous rather than built in.

**What the field does.** By (7), the sign of perceived agreement decides whether trust grows or decays. A receiver that adds $+p$ to $h_w$ reads its counterpart as more agreeable than it is and becomes more trusting: it is *tolerant*. A receiver that adds $-p$ manufactures disagreement that was not there and becomes more distrustful: it is *intolerant*. Throughout this paper $p > 0$ therefore denotes **tolerance towards in-group and intolerance towards out-group** emitters, and $p < 0$ the reverse: prejudice in favour of the other class. Figure 3 shows the displacement in the trust sector, showing that the field displaces the separatrix of the learning flow. What a population does with a displaced separatrix is the subject of §4.

## 4 Multi-agent systems

A single agent's error is $O(p)$ and local. This section asks what a population of them settles into: first with no bias at all, then over the plane of bias strength and prevalence, and finally what that plane implies for how such a system should be evaluated.

### 4.1 Emergent polarization

Nothing in §3.1 refers to an agent's class, so before adding any bias we ask what the population does unaided, i.e., for $p = 0$. It polarizes, spontaneously and into exactly two camps (Alves & Caticha, 2021). Two quantities per pair are enough to see it:

$$\rho_{IJ} \;=\; \frac{\hat{w}_I \cdot \hat{w}_J}{\|\hat{w}_I\|\,\|\hat{w}_J\|}, \qquad\qquad \eta_{J|I} \;=\; 1 - 2\Phi(h_\mu), \tag{9}$$

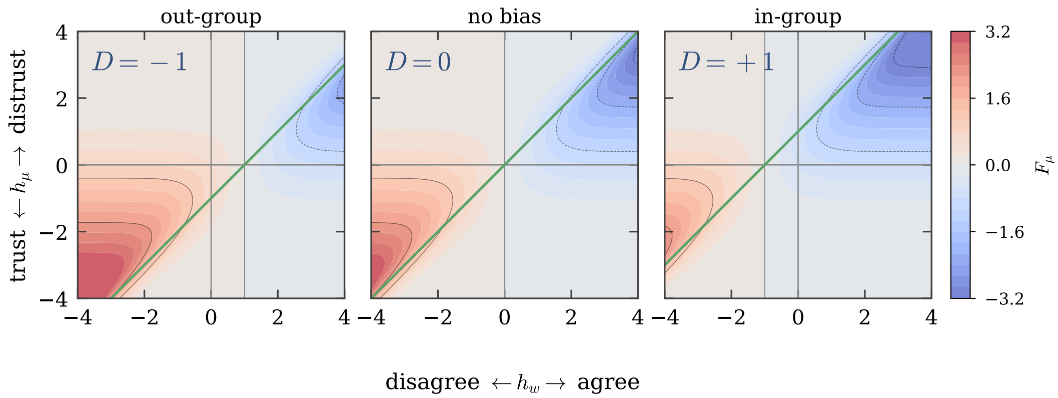



Figure 3: **The prejudice field displaces the separatrix.** $F_\mu$ over the plane of Figure 1, for a receiver that sits at $h_w$ but evaluates the modulation at $h_w + D$: the axes stay put and the green separatrix $h_\mu = h_w + D$ slides. Centre, no bias, and the two halves are balanced. Left, an out-group emitter under $p > 0$: the red region, where $F_\mu > 0$ and distrust grows, takes more of the plane. Right, an in-group emitter: the blue region, where trust grows, takes it instead. $F_\mu$ is the function to watch because the field enters through $h_w$, and it is the prefactor $1 - 2\Phi(h_w)$ in $F_\mu$ that carries it into the trust sector.

the *opinion overlap* of two agents and the *trust* a receiver $I$ places in an emitter $J$. The trust is the distrust field mapped onto a bounded, sign-carrying scale: $h_\mu$ is a distrust and $\eta$ is a trust.

**Order parameters.** Three numbers turn that into a statement about the population, each assembled from $\rho$ and $\eta$ alone. The first asks whether the alignment is real,

$$R_{w,\mu} = \left\langle \tfrac{1}{2}(\eta_{I|J} + \eta_{J|I})\, \rho_{IJ} \right\rangle, \tag{10}$$

averaged over unordered pairs and so lying in $[-1, 1]$, positive when agents trust those who agree with them. The other two answer what pairwise distributions cannot. The other two answer what pairwise distributions cannot. Every pair may have taken a side, but a bimodal histogram is equally consistent with three or four mutually hostile groups as with two, and telling them apart is a statement about triples rather than pairs (Heider, 1946; Cartwright & Harary, 1956). Two balance aggregates, one per sector, do it: they are positive when the disagreements in a population are mutually consistent, and they distinguish two opposed camps from any larger number of them. Appendix B.1 defines them. Like $\rho$ and $\eta$ they refer to no weight, gradient or embedding dimension, so nothing in (9) or (10) or in the aggregates depends on what the agents are made of, which is what will make them measurable on any population of interacting agents (§5).

**Where an unbiased population ends up.** Across eight independent populations of $N = 400$ agents at $p = 0$, $R_{w,\mu}$ rises from zero to $+0.56$, so the alignment between the sectors is not imposed anywhere in §3.1 but follows from the signs of the modulation functions. The camps are exactly two, and the balance aggregates say so sharply: of the 84 694 400 triples in these runs not one is unbalanced in trust (Appendix B.2). Two features of the split matter later. It is *spontaneous* — which agent lands in which faction is set by the initial condition, and the two are equal only up to fluctuation. And it is *class-blind*: the labels $\kappa_I$ exist, but no part of the dynamics has consulted them, so the factions cut across them. What §3.3 adds is not a mechanism for creating a split but one for aligning a pre-existing split with the labels.

**The agenda decides which sector polarizes first, and why.** Which sector leads reverses with the size of the agenda (Alves & Caticha, 2021). It is controlled by

$$\alpha = \frac{P}{K}, \tag{11}$$

the number of issues in play per degree of freedom available to take a position on them. Which side of 1 a real population sits on is not obvious: opinion is famously constrained, with stances on many

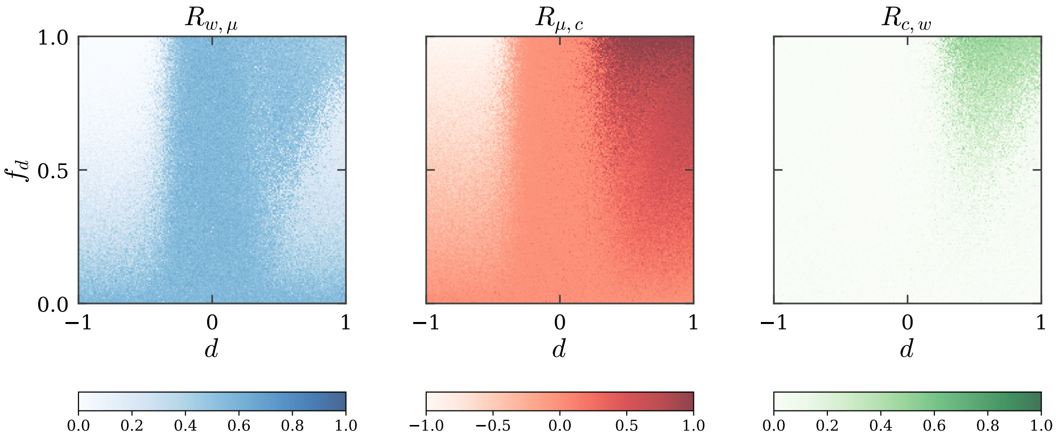



Figure 4: **The three correlations over the** $(p, f_p)$ **plane**, simple agenda ($\alpha = 0.17$). Each runs white to saturated across the whole of its range, $R_{\mu,c}$ included, so that these three panels are also the three colour channels of Figure 5. $R_{\mu,c}$ is the discrimination order parameter, changing sign at $p \approx 0$ ever more sharply as $f_p$ grows. $R_{c,w}$ is non-zero only for $p > 0$ and everywhere weaker than $R_{\mu,c}$: opinion follows class less readily than trust does, and how much less is what separates the two discriminatory phases. $R_{w,\mu}$ is high everywhere except at $p < 0$ above the quorum, where it collapses.

issues collapsing onto few latent dimensions, which argues for $\alpha > 1$; a domain in which positions are genuinely independent argues the other way. Both regimes give the same four phases of §4.2, so nothing below turns on the answer. Simple agendas polarize in trust first, complex agendas in opinion first, and the crossover is at $\alpha \approx 1.7$ (Appendix B.3).

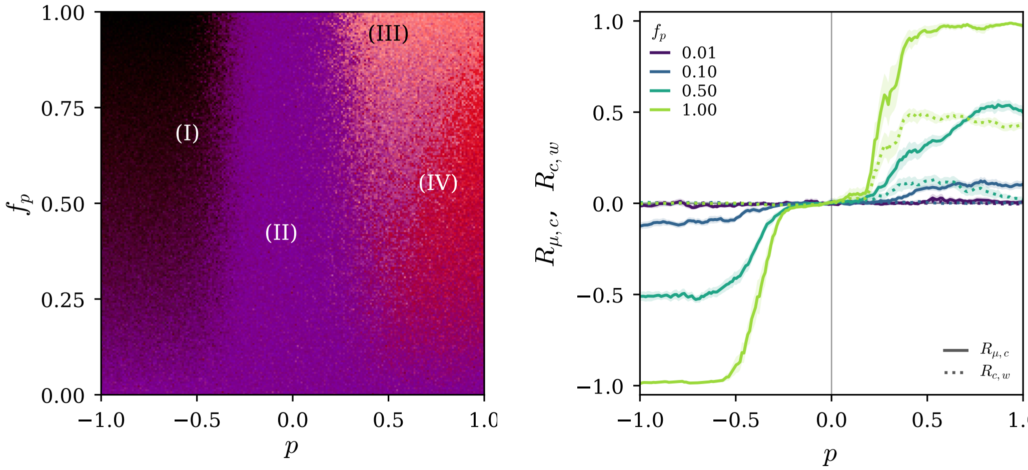
The asymmetry has an exact cause, which we can give. Every weight update is along $C_r\hat{x}$, which stays in the span of the agenda for all time, so the component of each agent's weight vector orthogonal to that span is conserved by the dynamics (Appendix A). For $\alpha < 1$ each agent keeps a frozen random component that no interaction can touch, which dilutes every overlap as $\rho_{IJ} \approx \pm f_I f_J$ and so caps how far opinion can align at all: $\langle|\rho|\rangle = 0.62$ at $\alpha = 0.17$ against 0.99 at $\alpha = 3.3$ (Appendix A). No amount of running closes that gap, because it is a property of the space and not of the time spent in it. Trust has no such ceiling: $\mu_{e|r}$ is one scalar per ordered pair rather than a projection of a high-dimensional vector, so $\eta$ saturates at any $\alpha$. The same argument accounts for the control reported alongside the effect, in which the reversal disappears when the covariance is held at $C = cI$: the ceiling requires not only a frozen component but an annealing one, since $C \to 0$ is what stops $\|u_I\|$ growing and leaves $f$ below one instead of creeping towards it. The same ratio decides, in §4.2, which of the two discriminatory phases a biased population ends up in; §5 takes up what the reversal implies for the question it is usually asked about.

### 4.2 The phase diagram

**Class order parameters.** Balance says whether a population is organized and (10) whether it is organized around agreement; neither can see whether it is organized around the label, because neither refers to one. Writing $G_{IJ} = \kappa_I \kappa_J$ for the class indicator (+1 for same-class pairs, −1 otherwise),

$$R_{\mu,c} = \left\langle \tfrac{1}{2} G_{IJ}(\eta_{I|J} + \eta_{J|I}) \right\rangle, \qquad R_{c,w} = \left\langle G_{IJ}\, \rho_{IJ} \right\rangle, \tag{12}$$

complete the set the phase diagram is built from, averaged over pairs and in $[-1, 1]$ like (10). $R_{\mu,c}$ is the order parameter of discrimination: it asks whether trust follows class, distinguishing a population that has merely polarized from one that has polarized *along the label*. $R_{c,w}$ asks whether opinion follows class, and separates the two discriminatory phases from each other: a population can have $R_{\mu,c}$ near one with $R_{c,w}$ near zero, and that combination is a distinct state rather than a degenerate case. Appendix J records why $G$ is signed rather than $\{0, 1\}$ and how the three correlations are normalized.




(a) The three maps of Figure 4 combined as the red, green and blue channels of one image: $R_{\mu,c}$ red, $R_{c,w}$ green, $R_{w,\mu}$ blue, each over the range its colour bar gives there, so a colour names a phase and a dark pixel is one where all three correlations are small.

(b) Both class correlations along $p$, one colour per $f_p$, pooling the rows of (a) in a narrow band around each with the standard error of the pooled mean shaded. Solid is $R_{\mu,c}$, dotted $R_{c,w}$. At $f_p = 0.01$ both are flat at zero however strong the bias, which is what makes region (II) a neutral state and not merely one with the label written into the other sector; as the fraction grows the response to $p$ steepens and saturates higher, trust ahead of opinion throughout.

Figure 5: **Four collective states, and what decides which one.** (I) dark, frustrated favouring of the out-group: trust follows the label and nothing else survives. (II) blue, opinion and trust correlated with each other but neither with class: neutral polarization. (III) pale, all three present, distrust and opinion both following the label. (IV) red, trust following the label without opinion: class alone. Region (II) runs along $p \approx 0$ at every $f_p$ *and* along small $f_p$ at every $p$, which (b) makes legible: the collective state turns on how many agents are biased as much as on how much each one is, and the fraction is not an attribute any single agent has. Simple agenda; the complex one has the same four states, with (III) taking a much larger share of the discriminatory half (Appendix C).

We simulate populations of $N = 40$ agents over a $200 \times 200$ grid in $(p, f_p)$, one population per grid point, and measure the five order parameters of §4.1 and (12) after a fixed number of interactions, with both classes prejudiced by (8). Parameters and their provenance are in Appendix K; each pixel is a single realization, so the speckle is population-to-population fluctuation, not a rendering artefact. Figure 4 maps the three correlations over that plane for the simple agenda, $\alpha = 0.17$; the complex agenda gives the same four states and is in Appendix C. Three correlations suffice to separate the four states; the balance aggregates add nothing to the classification over this plane and are mapped in Appendix B.4, where the frustrated corner is the one place they say something the correlations do not.

Composited into Figure 5, the three correlations name four collective states.

(I) **Frustrated favouring of the out-group.** For $p < 0$ above the quorum, $R_{\mu,c}$ is large and negative, which invites reading this half of the diagram as reverse discrimination. It is not, because nothing mirrors with it. A mirror image of the discriminatory state would need $R_{c,w}$ and $R_{w,\mu}$ to change sign along with $R_{\mu,c}$. Instead both go to zero, and that combination is what names the state: trust is strongly aligned with the label while uncorrelated with opinion, and $R_{w,\mu} \approx 0$ nowhere else in the diagram. The mirror state is not merely absent but unreachable. Suppose trust ran exactly against the label, every agent trusting the other class and distrusting its own, and take two agents of class $A$ and one of class $B$: both $A$s trust the $B$, so balance requires them to trust each other, and sharing a class is precisely what stops them. The triangle cannot close, no assignment of trust satisfies every triple at once, and the population settles into a spin glass rather than an inverted copy of the discriminatory state. The trust balance is the measurement that says so directly, going to $-0.71$ there where every correlation merely decays towards zero (Appendix B.4).

(II) **Neutral polarization.** At small $f_p$, and near $p = 0$ at any $f_p$, the society does exactly what §4.1 describes: it splits into two mutually distrustful camps, with $R_{w,\mu}$ large but $R_{\mu,c} \approx R_{c,w} \approx 0$, so the split cuts across the classes rather than along them. The bias is present in individual agents and absent from the population, because the prejudiced agents are outvoted by the class-blind majority they learn from. How far the region extends in $f_p$ is set by $p$. At full strength it is narrow, and $R_{\mu,c}$ simply tracks the fraction: $0.07$ at $f_p = 0.07$, $0.23$ at $0.20$, $0.97$ at $1$, each biased agent imposing its own alignment without needing the others. At moderate strength the region reaches far up the axis, because a lone biased agent is outvoted by the class-blind majority it learns from: at $p = 0.3$ a fifth of the population biased leaves $R_{\mu,c} = 0.05$, a tenth of the $0.59$ the same bias reaches when every agent carries it. Figure 5b shows both regimes as cuts: the widening gap between the $f_p$ curves at fixed $p$ is the fraction doing the work.

(III) **Distrust and disagreement both follow the label.** Past the quorum, at moderate $p > 0$, $R_{\mu,c}$ and $R_{c,w}$ are both large and comparable: the society still splits into two camps that distrust and disagree with each other, but the split now coincides with the class boundary. This is the state the mechanism of §3.3 predicts in full: the label has been written into both sectors at once.

(IV) **Class alone carries the hostility.** At moderate $f_p$ and strong $p$ the two class correlations come apart. $R_{\mu,c}$ is large and still growing with $p$, while $R_{c,w}$ has peaked and turned back, so $R_{c,w} \ll R_{\mu,c}$: trust follows the label and opinion barely does, and the agents distrust the other class without needing to disagree with it: the hostility is about the label and nothing else, with nothing left to argue about.

How the last two divide the discriminatory half is decided by the agenda. The ratio $R_{c,w}/R_{\mu,c}$ measures which of them a society is in, and it rises with $f_p$: opinion always lags trust in following the label, and closes the gap only when the prejudiced fraction is large. A narrow agenda keeps the ratio well below one everywhere, so it is in (IV) almost everywhere it is prejudiced at all and reaches (III) only as $f_p \to 1$. A broad agenda brings the ratio to one, passing through (IV) while the prejudiced fraction is still small and settling into (III) above it (Appendix C).

The reason is the conservation law. An agenda with $\alpha < 1$ leaves most of every opinion in directions no issue probes, and §4.1 showed that this caps $R_{c,w}$ however long the society runs; the prejudice field can align trust with class completely but opinion only within that ceiling. Region (IV) is where the ceiling binds. It is not the exclusive property of a narrow agenda (a broad one passes through it while the prejudiced minority is still small), but a narrow agenda cannot leave it.

Two features of Figure 5 are properties of the map from per-agent bias to collective state rather than of any single phase. A population with only a few per cent of biased agents is indistinguishable from neutral at any bias strength: at $f_p \leq 0.05$ and $p > 0.6$ we measure $R_{\mu,c} = 0.05$ against a realization-to-realization spread of $0.06$, and a null of $0.00 \pm 0.04$, so a single population there cannot be told from one with no bias in it at all. And the fraction and the strength do not act independently. At $p = 1$ the correlation is very nearly $f_p$ itself, the contributions of individual agents adding without interference; at $p = 0.3$ it is strongly convex in $f_p$, reaching $0.05$ at a fifth of the population and $0.59$ at all of it, so the biased agents accomplish little until enough of them agree (Figure 5b). §5 takes up what this implies for measuring such a population.

## 5 Discussion

**Optimality is context-dependent, and the context is group size.** The rule these agents use is optimal for the problem it was derived for: one student, one teacher, a noisy channel whose reliability must be inferred online. Nothing about it is careless. Yet the same rule, run in a large population with one class-correlated perturbation, converts an informationally empty label into a stable architecture of group distrust. The mechanism is specific: it acts through $F_\mu$, and a rule with no trust sector has none to bias: under a plain Hebbian update the prejudice field does nothing at all, since the update does not depend on $h_w$. What exposes the population is precisely the machinery that makes each agent good at inference, that it draws conclusions about *sources* as well as about the world. Giving interacting agents a well-founded model of each other's reliability is not a neutral improvement.

**No agent carries the fraction.** Per-agent bias does not determine the collective state. An audit that inspects agents one at a time returns $p$, or an estimate of it, for each agent, and that is not enough to place the population on Figure 5: the state turns on $f_p$ as well, and $f_p$ is not an attribute any single agent has. The standard protocol returns a verdict per agent and never forms the fraction, so a population can sit in a discriminatory phase with nothing in the audit's output that says so. Where the per-agent bias is also small enough to clear an individual tolerance, the audit passes every agent as well, and reports a population that is clean twice over. What such a protocol should measure instead is already fixed by (9): the order parameters that do determine the phase are built from a signed opinion alignment and a signed trust for each pair of agents and from nothing else — no weights, no gradients, no access to the embedding or its dimension, no assumption that the agents are perceptrons or that their update is the one in §3.1. Two elicited matrices per population therefore suffice to locate any interacting multi-agent system on the diagram, at a cost quadratic in the number of agents and independent of what the agents are made of. The claim is about measurement rather than deployment: we show that per-agent evaluation is blind to the collective state *in this model*, and that the order parameters which see it need nothing this model uniquely provides. Whether a population of language agents occupies phases of this kind we do not test; the instrument is cheap enough that it can be. Region (IV) is the case such an audit would find hardest to explain: the agents are not in conflict over any issue, so there is no disagreement for an inspection to find, and the hostility is carried by the label alone.

**A contested question with a parameter attached.** Whether groups come to dislike each other because they disagree, or come to disagree because they dislike each other, is argued over in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012). The reversal at $\alpha \approx 1.7$ makes it a question about a measurable quantity rather than a choice between theories: narrow agendas put affect first, broad ones put opinion first, and the two are regimes of one system rather than rival accounts. We do not claim this settles anything about any actual polity — the agents here are perceptrons — only that the dichotomy is not forced.

**Limitations.** The agents are single-layer perceptrons over a fixed shared embedding, all pairs may interact, and the society is closed and of fixed size. The dynamics anneals rather than reaching a stationary state, so our phases are phases at a finite interaction count, and the class labels are exactly inert, the clean version of a case that is never clean. Most importantly the result rests on one substrate, agents whose learning rule we can write down; whether the same phases appear where the agents are not analyzable is the obvious next question, which the order parameters of §4.1 were built to make answerable but which we do not answer. Human societies are a further step again.

## Reproducibility Statement

Every figure in this paper is produced by a script in the accompanying code, which contains the model, the order parameters, and the sweep driver, together with a test suite that checks the modulation functions against finite differences of $\log Z$, the invariants of the dynamics (the covariance stays symmetric positive-definite, the trust variance stays positive), and the order parameters against their known limits. One command regenerates all figures at a choice of three resolutions; sweeps are cached, so a figure can be restyled without re-simulating.

The model has few parameters and we state all of them, including the ones the formulation leaves open. §4.2 gives the values used; Appendix K gives the full table, separating quantities fixed by the model from those we chose, and records how each choice was made and what happens under alternatives. Two conventions where a literal reading of the model as previously written is ambiguous or inconsistent (the sign of the prejudice field, and the normalization of the class indicator) are resolved explicitly in Appendices I and J, with both readings implemented behind a flag so either can be reproduced.

## ETHICS STATEMENT

This paper studies how discriminatory structure can arise among interacting learning agents. Two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its "tolerance" is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. Where we discuss the trust-versus-opinion polarization literature, we do so to point out that the model identifies a parameter that selects between two regimes, not to adjudicate an empirical question about any actual polity.

The practical direction of the work is defensive. The result implies that a multi-agent system can occupy a discriminatory collective state while every agent in it passes an individual audit, and that the transition into that state is sharp in the strength of a per-agent bias. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here are inexpensive enough to serve as one.

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## A THE AGENDA'S ORTHOGONAL COMPLEMENT IS CONSERVED

Let $V_P = \mathrm{span}\{\hat{x}_1, \ldots, \hat{x}_P\}$ be the subspace spanned by the agenda. *Claim:* for every agent, the component of $\hat{w}$ orthogonal to $V_P$ is unchanged by (4).

By (4) the weight update is a multiple of $C_r\hat{x}$ with $\hat{x} \in V_P$, so it is enough that $C\hat{x} \in V_P$ at all times. At initialization $C = I$ and $C\hat{x} = \hat{x} \in V_P$. Suppose $C_t\hat{x}_i \in V_P$ for every issue. The covariance update is $C_{t+1} = C_t + a\,(C_t\hat{x})(C_t\hat{x})^\top$ for the issue $\hat{x}$ just seen, so for any issue $\hat{y}$,

$$C_{t+1}\hat{y} \;=\; C_t\hat{y} \;+\; a\,(C_t\hat{x})\left(C_t\hat{x}\cdot\hat{y}\right), \tag{13}$$

in which the first term lies in $V_P$ by hypothesis and the second is a multiple of $C_t\hat{x} \in V_P$. Hence $C_t\hat{x}_i \in V_P$ for all $t$ by induction, every increment to $\hat{w}$ lies in $V_P$, and the orthogonal component is conserved. □

The conservation is exact in the simulations: in the run of Figure 6, $\|w_\perp\| = 4.956$ at initialization and $4.956$ after $780\,000$ interactions, and the test suite asserts it numerically. It is a property of the *agenda*, not of the prejudice field: it holds at every $p$ and $f_p$.

**The ceiling it puts on opinion overlap.** Split $w_I = u_I + n_I$ into the learned part inside $V_P$ and the frozen part outside it. Learning aligns the $u_I$ while the $n_I$ stay random and mutually almost orthogonal, so

$$\rho_{IJ} \;\approx\; \pm f_I f_J, \qquad f_I = \frac{\|u_I\|}{\|w_I\|}, \tag{14}$$

and opinion overlap is diluted by whatever fraction of each weight vector learning cannot reach. Two facts fix that fraction. For $\alpha < 1$ a $(1-\alpha)$ fraction of directions is untouched, so $n_I \neq 0$; and because the dynamics anneals ($C \to 0$, so $\|u_I\|$ stops growing), $f$ settles below one instead of creeping towards it. Measured across four orders of magnitude in $\alpha$:

| $\alpha = P/K$ | 0.03 | 0.17 | 0.50 | 1 | 3.3 |
|---|---|---|---|---|---|
| $f^2$, predicted by (14) | 0.29 | 0.62 | 0.84 | 1.00 | 1.00 |
| $\langle\|\rho\|\rangle$, measured | 0.30 | 0.62 | 0.82 | 0.93 | 0.99 |

For $\alpha < 1$ the prediction holds to 0.02. For $\alpha \geq 1$ the agenda spans $\mathbb{R}^K$, the complement is empty, and the ceiling disappears, the residual shortfall being the finite number of interactions rather than a conserved quantity.

# B The balance order parameters

Balance over triples is what separates a population organized into opposing camps from one in disorder, and it is the only diagnostic here that can count the camps. This appendix defines the two aggregates, reports what they do in an unbiased population, gives the agenda effect they make visible, maps them over the $(p, f_p)$ plane, and gives the identity that makes them cheap to compute.

## B.1 Definition

Every pair may have taken a side, but a pair can disagree in a population where everyone disagrees with everyone, and a bimodal histogram is equally consistent with three or four mutually hostile groups. What separates organized disagreement from disorder is whether the disagreements are mutually consistent, which is a statement about triples (Heider, 1946; Cartwright & Harary, 1956). A triple is balanced in opinion when $\rho_{IJK} = \rho_{IJ}\rho_{JK}\rho_{KI} > 0$, and balanced in trust when the same holds for trust averaged over the two directions round the triangle, $\eta_{IJK} = \frac{1}{2}(\eta_{J|I}\eta_{K|J}\eta_{I|K} + \eta_{I|J}\eta_{J|K}\eta_{I|K}) > 0$. The aggregates

$$B_\rho = \langle \rho_{IJK} \rangle_{(IJK)}, \qquad B_\eta = \langle \eta_{IJK} \rangle_{(IJK)} \tag{15}$$

run over all $\binom{N}{3}$ triples (Alves & Caticha, 2021). A random population has $B_\rho \approx B_\eta \approx 0$; two internally coherent, mutually opposed blocs give $B_\rho = B_\eta = 1$ whatever the blocs are made of; and a population that cannot settle stays near zero or goes negative. Like $\rho$ and $\eta$ themselves they refer to no weight, gradient or embedding dimension.

## B.2 An unbiased population, and why the camps are exactly two

Across eight independent populations of $N = 400$ agents at $p = 0$, trust balance reaches $B_\eta = 0.96$ while opinion balance reaches only $B_\rho = 0.19$, the gap §B.3 explains. That the camps are exactly *two* follows from the same triples, since a signed complete graph is balanced precisely when it splits into two mutually hostile cliques (Cartwright & Harary, 1956): of the 84 694 400 triples in these runs, not one is unbalanced in trust, and in opinion the shortfall is 1 590, two thousandths of one per cent, traceable to the few pairs whose overlap is so close to zero that its sign is arbitrary.

Figure 6 follows both pairwise distributions of (9) through an unbiased run: opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at $\pm 1$, so the split in trust is built by the dynamics and not supplied by the initial condition. Figure 7 follows the two balance aggregates through the $(B_\rho, B_\eta)$ plane as the agenda varies. Both effects are reported by Alves & Caticha (2021); §4.1 gives the numbers, and the conservation law of Appendix A accounts for the second.

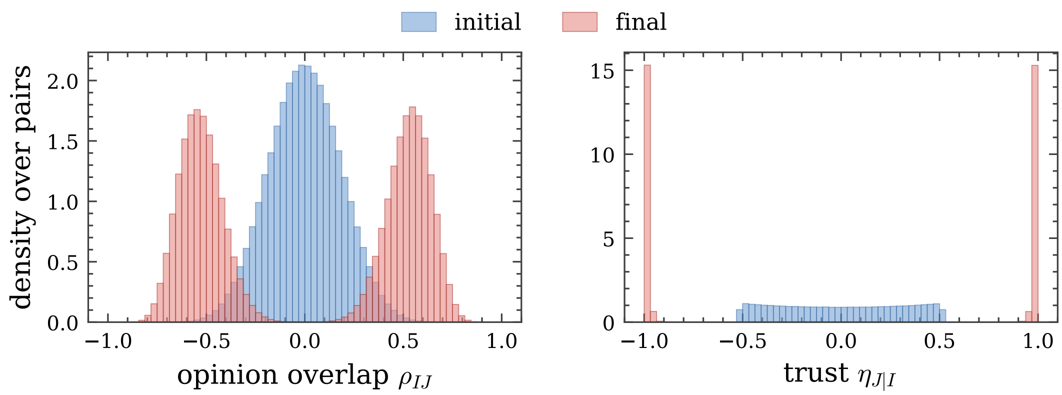



Figure 6: **An unbiased society splits in two.** Eight independent societies of $N = 400$ agents at $p = 0$, simple agenda, before and after the calibrated 500 interactions per ordered pair, with their pairs pooled: 638 400 unordered pairs for the opinion overlap and 1 276 800 ordered pairs for the trust. Blue is the initial state, red the final one. Neither panel depends on how the agents are labelled, so neither can be an artefact of sorting them. Opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at $\pm 1$, so the split in trust is built by the dynamics and not supplied by the initial condition. Bimodality alone would show only that every pair has taken a side; that the sides are exactly *two* blocs is the sign-balance figure quoted in the text.

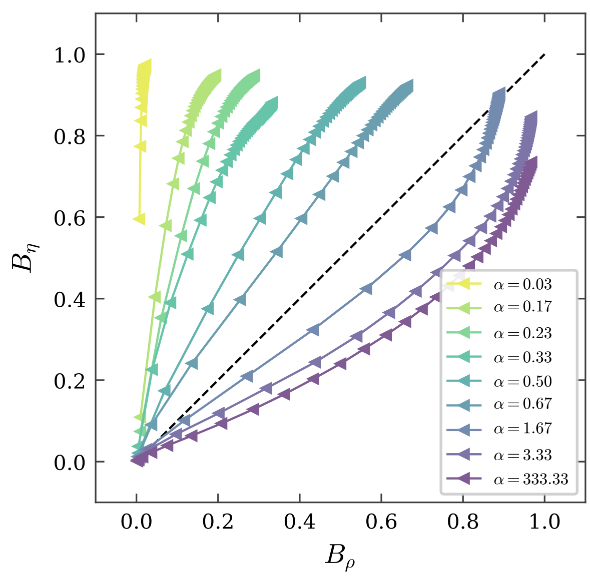



Figure 7: **Which polarization comes first is set by the agenda.** Trajectories in the $(B_\rho, B_\eta)$ plane with no prejudice ($p = 0$), one curve per agenda complexity $\alpha = P/K$, markers equally spaced in time; they crowd at the end because the annealed dynamics slows, as §3.1 requires. Simple agendas run above the diagonal (distrust forms first), complex ones below it (disagreement first), crossing over at $\alpha \approx 1.7$.

### B.3 The agenda effect

Which sector polarizes first reverses with the size of the agenda (Alves & Caticha, 2021). Following a population through the $(B_\rho, B_\eta)$ plane rather than measuring it once at the end separates the two: simple agendas run above the diagonal, ending at $(B_\rho, B_\eta) = (0.02, 0.97)$ for $\alpha = 0.03$; complex ones run below it, ending at $(0.97, 0.74)$ for $\alpha = 333$. The crossover is at $\alpha \approx 1.7$. The ceiling of

Appendix A is what caps $B_\rho$ for a narrow agenda, and it is directly legible here: $B_\rho$ stays near $0.1$ for a narrow agenda and saturates for a broad one.

### B.4 Over the $(p, f_p)$ plane, and the frustrated corner

$B_\rho$ and $B_\eta$ are not needed to separate the four states, and over the $(p, f_p)$ plane they are close to redundant with $R_{w,\mu}$: the Pearson correlation across all $40\,000$ grid points is $0.97$ and $0.98$ for $B_\eta$ and $0.98$ and $0.99$ for $B_\rho$, at $\alpha = 0.17$ and $\alpha = 3.3$ respectively. Their maps therefore look like the $R_{w,\mu}$ panel of Figure 4, and that is why §4.2 does not show them.

They are not identical to it, and the difference is worth recording even though §4.2 does not lean on it. $R_{w,\mu}$ never falls appreciably below zero (its minimum over either sweep is $-0.03$), so in the frustrated region it merely decays towards zero. $B_\eta$ goes to $-0.71$, and a negative balance is a stronger statement than a small positive one: it says the trust relations cannot all be satisfied at once, which is what frustration means. Where the correlations locate the state, the balance names the mechanism. $B_\rho$ plays the same role for the other sector, being the quantity in which the conservation-law ceiling of Appendix A is directly legible: it stays near $0.1$ for a narrow agenda and saturates for a broad one.

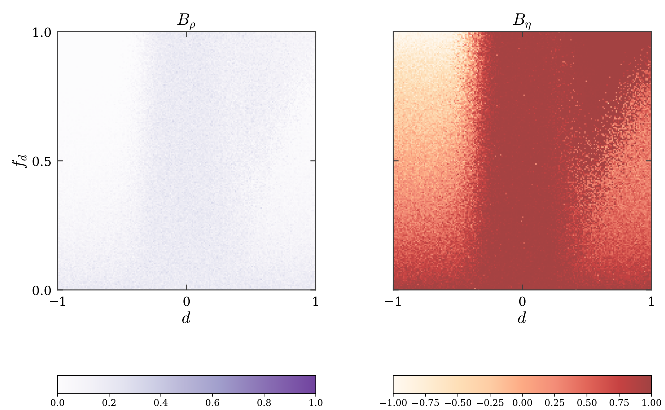



Figure 8: **The two balance aggregates**, simple agenda ($\alpha = 0.17$). $B_\eta$ is the only measurement in the paper in which the frustrated region announces itself, going strongly negative at upper left where every correlation merely goes to zero. $B_\rho$ stays small everywhere.

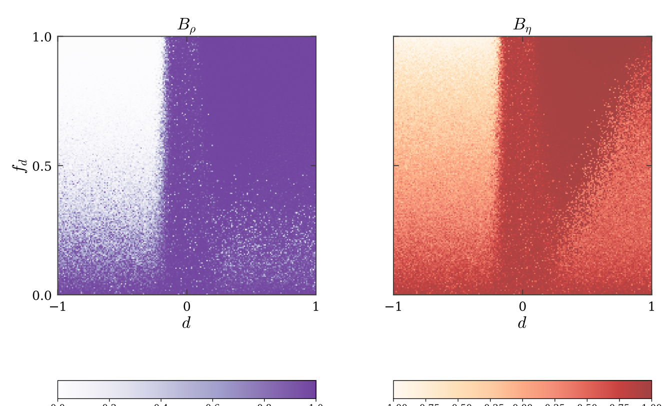



Figure 9: **The two balance aggregates**, complex agenda ($\alpha = 3.3$). $B_\rho$ saturates rather than staying near zero, which is what removes the ceiling on $R_{c,w}$. $B_\eta$ is essentially unchanged, including the anti-balanced corner at upper left.

### B.5 Closed form

Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a phase diagram. For any matrix $M$ with unit diagonal, either $\rho$ or $\eta$,

$$\sum_{\substack{(I,J,K) \\ \text{distinct, ordered}}} M_{IJ} M_{JK} M_{KI} \;=\; \mathrm{tr}(M^3) - 3\Big(\textstyle\sum_{I,J} M_{IJ} M_{JI} - N\Big) - N, \tag{16}$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse, so the triple average is $\big[\text{right-hand side}\big] / \big(6\binom{N}{3}\big)$. For symmetric $M$ this is $B_\rho$; for the asymmetric trust matrix the two cycle orientations are exactly the two terms of $\eta_{IJK}$, so the same expression gives $B_\eta$. Both are batched matrix products. The identity is verified against enumeration in the test suite.

## C The complex agenda

§4.2 maps the $(p, f_p)$ plane for a simple agenda, $\alpha = 0.17$. The complex agenda, $\alpha = 3.3$, behaves the same way qualitatively: the same four states, in the same arrangement, with the same dependence on $f_p$ and the same sign change in $p$. What differs is quantitative, and it is the agenda-size effect of §4.1 showing up in the phase diagram. Opinion is no longer capped, so $R_{c,w}$ rises to meet $R_{\mu,c}$ instead of lagging it, and region (III) takes most of the discriminatory half rather than a sliver at the top of it. $B_\rho$ saturates at $0.98$ across the whole $p > 0$ region rather than staying near $0.1$.

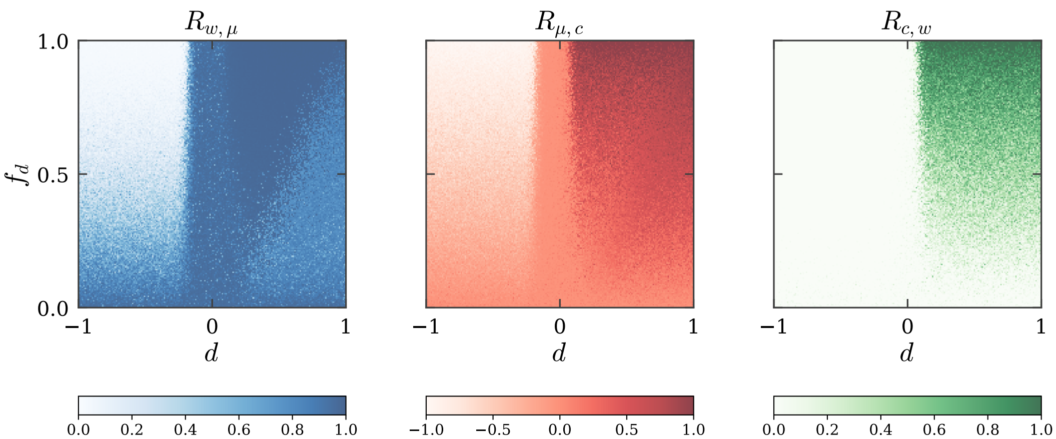


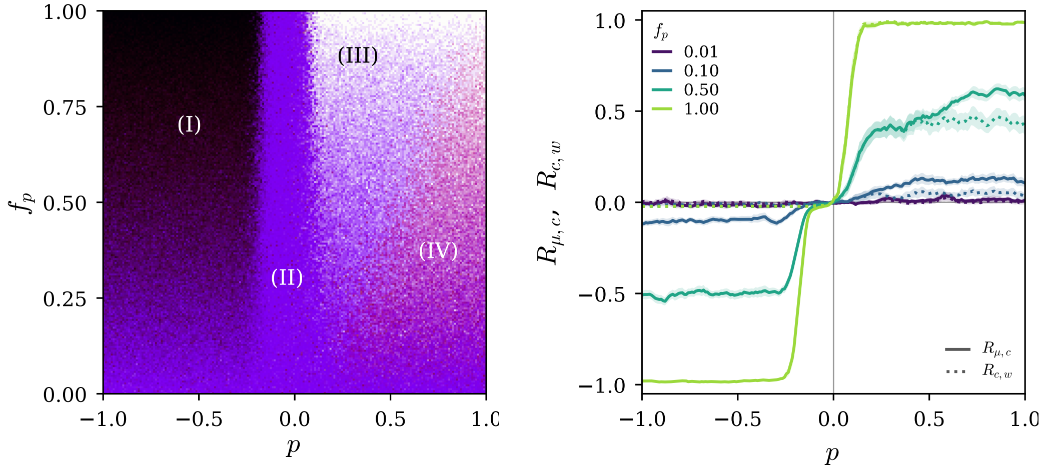

Figure 10: **The three correlations**, complex agenda ($\alpha = 3.3$). Figure 4 for the other agenda size. $R_{c,w}$ is the panel that changes: it now tracks $R_{\mu,c}$ over most of the $p > 0$ half instead of remaining a fraction of it.

## D All four modulation functions

## E Modulation function for LLM in-context learning

§3.2 states what the measurement finds. This appendix gives the design it rests on, the two ways of reading the receiver's conviction, and where the instrument runs out.

**Design.** Twenty open, low-valence themes, each carrying an opinion the agent wrote itself and a set of further opinions it rated against that one. A condition is a theme, an in-context history of previous statements by another agent (from which $k$, agreements minus disagreements, indexes the columns), and a framing that says the agent holds its own view firmly or tentatively. The history is the whole of the learning: it is the only thing that distinguishes a trusting condition from a distrusting one, it is presented as ordinary dialogue, and no weight is changed anywhere. Every question asked




(a) All four states are present. Region (III) now occupies the upper part of the discriminatory half, with (IV) confined to moderate $f_p$, where the prejudiced fraction is still small enough that opinion has not caught up with trust.

(b) Both class correlations along $p$, one colour per $f_p$, pooling the rows of (a) in a narrow band around each with the standard error of the pooled mean shaded. Solid is $R_{\mu,c}$, dotted $R_{c,w}$. At $f_p = 0.01$ both are flat at zero however strong the bias, which is what makes region (II) a neutral state and not merely one with the label written into the other sector; as the fraction grows the response to $p$ steepens and saturates higher, trust ahead of opinion throughout.

Figure 11: **The phase diagram and its line cut**, complex agenda ($\alpha = 3.3$): Figure 5 for the other agenda size. The $f_p$ dependence is unchanged — flat at zero at $f_p = 1\%$ however strong the bias, steepening and saturating higher as the fraction grows — while $R_{c,w}$ now tracks $R_{\mu,c}$ instead of lagging it. Broadening the agenda changes which discriminatory state the population lands in, not whether the fraction of biased agents governs the approach to it.

is about the *other* agent, and there are three of them. *Reliability*: the chance, from 0 to 100, that the other agent's judgement on this theme is sound. Since the model takes the probability of the emitter conveying the wrong label to be $\Phi(h_\mu)$, this is $\Phi(-h_\mu)$, so $h_\mu = -\Phi^{-1}(r)$. *Agreement*: the chance their next statement is one the receiver would agree with. Writing $c = \Phi(|h_w|)$ and $e = \Phi(h_\mu)$, the two agree when both are right or both are wrong, so this answer is $c + e - 2ce$, which is (3) on the branch where the message agrees. The magnitude is what is asked for and all that is defined: the question precedes any message, so there is no $\sigma_e$ to sign $h_w$ with, and $c$ is a property of the receiver alone. Given $e$ it inverts for that conviction, $c = (q - e)/(1 - 2e)$, and the message supplies the sign when it arrives. *Conviction*: the same agreement question, asked about a third party the receiver is told it holds to be sound. For such a source $e \to 0$ and $c + e - 2ce$ collapses to $c$, so this reads the conviction off directly, with no quotient to be singular and no range for the answer to fall outside, and it is what places a condition along the $h_w$ axis. The reliability question is then asked again with one message added and nothing else changed, and the difference of the two answers is $\Delta h_\mu$. The message is drawn from the theme's strongly agreeing or strongly disagreeing column, and supplies only the sign of $h_w$, as the emitter does in the model.

Prior trust is never asserted, only learned in context from the history, so there is no stated level for the agent to repeat back and the before/after pair is a difference of two answers to one question. Nothing asks the agent how strongly it believes its own opinion: an agent told that a view is its own reads holding it as a commitment, and answers such a question at the ceiling or by stepping outside the persona, which are different quantities. Conviction is recovered by the questions above instead. The run is 356 conditions and 6 764 calls to `gpt-5.6-luna`, cached by request hash.

**What reproduces.** Figure 2a puts the measurement on the plane of Figure 1. The sign of $\Delta h_\mu$ follows the sign of the agreement in 95.2% and 84.8% of conditions for agreeing and disagreeing

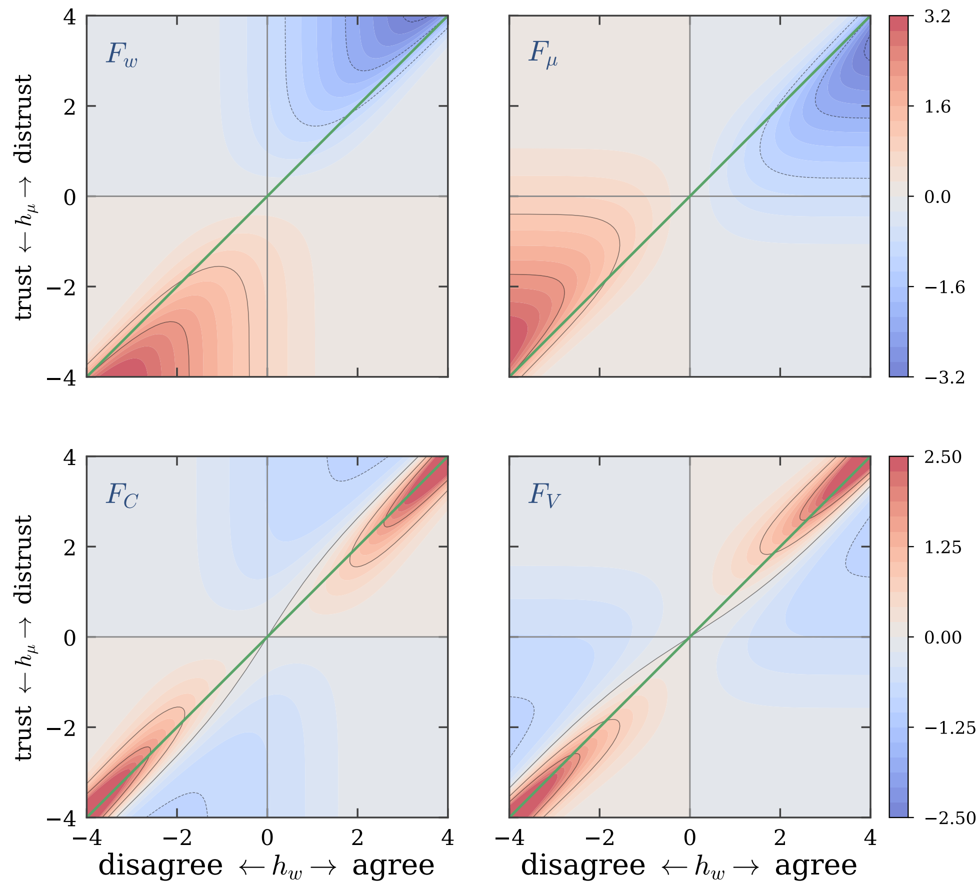



Figure 12: The two functions that move the means, $F_w$ and $F_\mu$, above (the pair shown in Figure 1) and the two that anneal the uncertainties, $F_C$ and $F_V$, below. Each row shares a range. The mirror symmetry across $h_\mu = h_w$ holds in both rows: $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. $F_C$ and $F_V$ are negative almost everywhere, which is why both uncertainties shrink monotonically and the dynamics anneals (§3.1).

messages; its magnitude is $1.97$ against $0.50$ (agreeing message, dissonant against consonant half) and $1.69$ against $0.34$ (disagreeing message), so surprise gates the size of the update by four to five while agreement fixes its direction. Over all 712 conditions it correlates with $F_\mu$ at $r = 0.85$ with $86.0\%$ sign agreement, under one fitted positive scale. Told to hold its opinion firmly rather than tentatively, the agent updates its trust more ($|\Delta h_\mu| = 1.34$ against $0.90$), the ordering $|1 - 2\Phi(h_w)|$ predicts.

Figure 2b is the same measurement as a function of prior trust alone, where the theory carries one fitted scale and one fitted conviction. That conviction is a lower bound, $|h_w| \gtrsim 2.9$: past that value the residual is flat, because the peak of $|F_\mu|$ moves outward with conviction and, in the limit, $F_\mu$ becomes the Gaussian hazard rate and has no peak at all. At the fitted value the peak sits at $h_\mu = +2.29$, outside the $\pm 2.05$ the readout can reach, so what the measurement covers is the monotone flank and the turnover the model also predicts is not observable here in either direction.

**Two ways to read the conviction, and why we use this one.** The $h_w$ coordinate could equally come from the inversion $c = (q - e)/(1 - 2e)$, and reporting both matters because they do not agree ($r = 0.46$ on the 380 conditions where both are defined). The inversion is admissible only where $q$ lies in $[r, 1-r]$, and against a distrusted interlocutor that band is narrow: the agents routinely predict disagreement more strongly than $c = 1$ permits, so the condition is censored rather than measured. It survives on 280 of 712 conditions, and the censoring falls hardest in the distrusted half — exactly where the plane is most informative. Making the histories more hostile does not widen the band, because the stated reliability floors near 0.24 and in fact *rises* with a longer disagreeing history, from 0.236 at two contradicting statements to 0.341 at five. The direct read needs no such filter and is defined on all 712 conditions, so it is the one plotted.

**Where the measurement runs out.** The direct conviction question asks what a sound third party would say, which a receiver may answer from how common it takes its view to be rather than from how sure it is; perceived consensus and conviction are not separated by this instrument. Both conviction readings are also compressed relative to the curve fit — median $|h_w|$ is 0.67 direct and 1.15 by inversion, against the $\gtrsim$ 2.9 Figure 2b prefers — which leaves signs intact and makes Figure 2a conservative rather than wrong. And the readout is a stated probability, clipped before the probit, so $|h_\mu|$ cannot exceed 2.05 and the deep corners of Figure 1 are out of reach in every condition.

What this is evidence for is therefore narrow and worth stating plainly. A large language model learning in context, asked only what it expects a source to say and how sound it finds them, revises its trust with the sign, the gating and the conviction-ordering that (7) derives from the inference problem, in a setting where no weight moves and nothing is shared with the learning rule of §3.1. That is evidence about the problem rather than the mechanism: it does not show that in-context learning implements (4)–(5), and the correspondence is qualitative, resting on a fitted positive scale and holding only inside the window the readout reaches.

## F Derivation of the learning algorithm

The receiver's belief before an interaction is the product of Gaussians (1). It observes $\sigma_e$, the emitter's opinion on issue $\hat{x}$, and treats it as a report of an unknown true label $\sigma_T$ transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalizing the joint distribution of labels and of the issue as the emitter may have parsed it, $y_t$,

$$L(\sigma_e \mid \hat{x}, u) \;=\; \sum_{\sigma_T} \int P(\sigma_e \mid \sigma_T, y_t, \hat{x}, u)\, P(\sigma_T \mid y_t, \hat{x}, u)\, P(y_t \mid \hat{x}, u)\, \mathrm{d}y_t, \tag{17}$$

with $P(\sigma_e \mid \sigma_T, z) = (1-\varepsilon(z))\delta_{\sigma_e,\sigma_T} + \varepsilon(z)\delta_{\sigma_e,-\sigma_T}$ and $P(\sigma_T \mid y_t, w) = \Theta(\sigma_T\, y_t \cdot w)$. Two modelling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with $z$ Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the trust sector; a Beta distribution for $\varepsilon$, the tempting alternative, does not. Taking $P(y_t \mid \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e \mid z, \hat{x}, w) \;=\; \Phi(z)\, \Phi(-w \cdot \hat{x}\sigma_e) + \big(1 - \Phi(z)\big)\, \Phi(w \cdot \hat{x}\sigma_e), \tag{18}$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e \mid \hat{x}, D_t) \;=\; \mathbb{E}_{Q_t}\big[L(\sigma_e \mid z, u, \hat{x})\big] \;=\; \sum_{\sigma_T} \mathbb{E}_z\big[P(\sigma_e \mid \sigma_T, \hat{x}, z)\big]\, \mathbb{E}_w\big[\Phi(\sigma_T\, w \cdot \hat{x})\big], \tag{19}$$

and the Gaussian integrals collapse onto the two scaled fields (2), giving (3): the evidence depends on the receiver's state only through $(h_w, h_\mu)$. This is the symmetry that makes the two sectors mirror images.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t \frac{\partial \ln Z}{\partial \hat{w}}, \qquad C_{t+1} = C_t + C_t \frac{\partial^2 \ln Z}{\partial \hat{w}\, \partial \hat{w}^\top} C_t, \tag{20}$$

and identically in the trust sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields (4)–(5) with the modulation functions (7). The first is a Hebbian term $\sigma_e \hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

## G Genealogy of the learning rule

The update (4)–(5) descends from the variational optimization of modulated Hebbian on-line learning (Kinouchi & Caticha, 1992; Kinzel & Ruján, 1990), given a Bayesian reading by Opper (1996; 1998) and applied across architectures by Solla & Winther (1999); the entropic-dynamics formulation we use is Caticha (2020). Learning through a noisy channel whose noise level must itself be inferred was analyzed by Kinouchi & Caticha (1993) and Biehl et al. (1995); Alves & Caticha (2016) identified that inferred level with distrust and made it dynamical, with applications to voting (Caticha et al., 2015) and to panels of appellate judges (Caticha & Alves, 2019), and a mean-field treatment by Simões & Caticha (2018). The blame-attribution behaviour resembles the least-action rule of Mitchison & Durbin (1989), which suffers retarded learning (Kabashima, 1994), whereas the optimized algorithm attains generalization error decaying as $\alpha^{-1}$ even under output noise (Simonetti & Caticha, 1996).

## H The prejudice field

Equation (8) is one filling of a $2 \times 2$ matrix $D$ indexed by the classes of receiver and emitter, $h_w^p = h_w + D_{\kappa_r,\kappa_e}$. Four numbers are free. Since $\kappa_r$ and $\kappa_e$ each take two values, any filling decomposes uniquely into four orthogonal components,

$$D_{\kappa_r\kappa_e} \;=\; a \;+\; b\,\kappa_r \;+\; c\,\kappa_e \;+\; p\,\kappa_r\kappa_e, \tag{21}$$

with independent magnitudes and one meaning each. $a$ shifts perceived agreement by the same amount whoever is speaking: a uniform credulity, referring to no label. $b$ depends only on who is *listening*, so one class is systematically the more credulous. $c$ depends only on who is *speaking*, so one class is believed more by everyone, its own members included. $p$ depends only on whether the two classes *match*, which is in-group favouritism and out-group hostility at once, and is what (8) uses with $a = b = c = 0$.

Relabelling the classes sends $(a, b, c, p) \to (a, -b, -c, p)$. Only $a$ and $p$ survive it, and $a$ does not refer to the label, so $p$ is the one component that is both class-dependent and symmetric between the classes. That is why the main text uses it: under $p$ alone the two classes are interchangeable, so any asymmetry in the outcome is spontaneous rather than imposed. The components that change sign, $b$ and $c$, name a class rather than a relation between classes.

**None of the four is redundant.** It is tempting to read $a$ as a shift that the dynamics could absorb, which would make "both classes favour their own" ($a = p = \frac{1}{2}$) and "both are hostile to the other" ($a = -\frac{1}{2}, p = \frac{1}{2}$) the same experiment. They are not. The updates carry no free scale: $\Phi$ and $g$ have unit width, so $h_w = 1$ means one standard deviation of the receiver's belief, and multiplying both fields by a constant changes the ratio $F_\mu/F_w$ and so the direction of the flow, not merely its speed. Nor is a shift absorbable, because $h_w$ already carries the emitter's sign $\sigma_e$: adding a constant to it is not a bias on the perceptron, which would flip with $\sigma_e$, but a standing over- or under-estimate of agreement that does not.

What a shift does is act on one sector only. The sign of $F_w$ is fixed by $1 - 2\Phi(h_\mu)$, which contains no $h_w$, so $a$ cannot move the separatrix of the opinion sector. The sign of $F_\mu$ is fixed by $1-2\Phi(h_w)$, so $a$ moves the trust separatrix to $h_w = -a$: with $a > 0$ a receiver builds trust even in an emitter it mildly disagrees with, and the fraction of the plane in which distrust grows falls from $0.499$ at $a = 0$ to $0.437$ at $a = \frac{1}{2}$. A uniform credulity therefore changes how readily trust forms at all, and will move the phase boundaries of §4.2, while remaining unable to correlate anything with the label.

**The four components pair with four ways trust can carry a label.** The trust of (9) is a *directed* matrix: $\eta_{J|I}$, what receiver $I$ makes of emitter $J$, need not equal $\eta_{I|J}$. Its class structure therefore decomposes exactly as the field does, into four channels averaged over ordered pairs $I \neq J$,

$$T_\mu = \langle \eta_{J|I} \rangle, \quad R_{\mu,\mathrm{cred}} = \langle \kappa_I\,\eta_{J|I} \rangle, \quad R_{\mu,\mathrm{stat}} = \langle \kappa_J\,\eta_{J|I} \rangle, \quad R_{\mu,\mathrm{c}} = \langle \kappa_I\kappa_J\,\eta_{J|I} \rangle, \tag{22}$$

one for each component of (21) and orthogonal for the same reason. The last is $R_{\mu,c}$ of (12) exactly, not an analogue of it: $\kappa_I\kappa_J$ is unchanged by exchanging the pair, so averaging the directed product over ordered pairs is the same as averaging the symmetrized trust over unordered ones. The other three have no counterpart in §4.2, and $R_{\mu,\mathrm{stat}}$ is the one to watch, being the correlation between an agent's class and the trust it *receives*: prestige, or its negative, stigma.

The omission is not a matter of degree. *A field in $b$ or $c$ alone leaves every order parameter of this paper at zero*, and for classes of equal size the statement is exact. Each of (10) and (12) contracts trust with a weight that is even under relabelling, $\kappa_I\kappa_J$ or no class weight at all, whereas $b$ and $c$ write trust into a class-odd structure; and a class-odd structure vanishes against a class-even weight because $\sum_I \kappa_I = 0$. Concretely, under a pure $c$ field trust depends on the emitter alone, $\eta_{J|I} = s(\kappa_J)$, so the symmetrized trust of a pair is $\frac{1}{2}\big(s(\kappa_I) + s(\kappa_J)\big)$: this is $+s$ on same-class pairs of one class, $-s$ on those of the other and 0 on mixed pairs, against a class indicator of $+1, +1, -1$. The two same-class terms cancel term by term. The balance aggregates vanish for a prettier reason. In the cycle $\eta_{J|I}\eta_{K|J}\eta_{I|K}$ each of the three agents appears exactly once as the *emitter*, so under a pure $c$ field the sign of the product is $(-1)^{n_B}$ with $n_B$ the number of agents of the disfavoured class in the triple; balance is decided by the parity of how many stigmatized members a triple has, and equal class sizes make the two parities equinumerous. So $B_\eta$ under a status field is not small but identically zero, while the population is as ordered as it can be.

**$b$ and $c$ are transposes, and nothing here can tell them apart.** The matrix $D$ of a pure $b$ field is the transpose of that of a pure $c$ field, and since $D_{\kappa_I\kappa_J}$ enters the update of $\mu_{J|I}$ in place, the trust matrices they produce are transposes too. Every quantity this paper measures is invariant under transposing $\eta$: $R_{w,\mu}$ and $R_{\mu,c}$ use only the symmetrized trust, the balance aggregates average a cycle against its reverse, and $R_{c,w}$ does not involve $\eta$ at all. Of (22) only $R_{\mu,\mathrm{stat}}$ and $R_{\mu,\mathrm{cred}}$ change, and they exchange values. Yet the two states are opposites. Under $c$ one class is disbelieved by everyone, its own members included; under $b$ one class believes nobody while the other believes everyone. A population that is stigmatized and a population that is suspicious are as different as those words, and neither $R_{\mu,c}$ nor the balances separate them, nor either from a population with no bias at all. Keeping the elicited trust matrix directed, and contracting it against each of the four class weights rather than the matching one alone, costs nothing beyond what §5 already asks for, and this is the case that shows why it is worth doing.

**What we have not run.** The main text varies $p$ with $a = b = c = 0$ throughout, and the two statements above are about what the order parameters would report rather than about what a population does. We have not simulated the other three components here, and say nothing about the collective states they produce: $c$ in particular describes an advantage attached to a class rather than to a relationship, credited by both classes alike, and what a population does with it is a separate question from the one Figure 5 answers.

## I The sign of the prejudice field

The sign of the prejudice field is fixed by the algorithm and not free. By (7), $F_\mu = (1 - 2\Phi(h_w))g(h_\mu)/Z$ is negative when $h_w > 0$, so perceived agreement drives distrust $\mu$ *down*. A receiver that adds $+p$ to $h_w$ therefore becomes more trusting (tolerant) and one that adds $-p$ becomes more distrustful (intolerant). Case 6 with $p > 0$ consequently gives in-group trust and out-group distrust, i.e. the discriminatory phase, which is the convention used throughout.

The companion manuscript (Anonymous, 2026) tabulates the in-group entry as $-p$ while its text and figures place the discriminatory phases at $p > 0$; those two statements are inconsistent, and taken literally the tabulated signs put the discriminatory phases at $p < 0$, mirroring every map in $p$. The same inconsistency appears in its description of the learning flow, where the panel labelled $D_{e|r} = -p$ is called the more tolerant one although $-p$ is the intolerant entry, and where the "distrust and disagree" basin is said to grow for $p > 0$ although adding $+p$ shrinks it. A single global sign reconciles everything; we adopt it in the form that leaves (8) untouched. The accompanying code implements both readings behind a flag, and Figure 13 runs the same sweep under each: the two rows are mirror images in $p$, and only the first matches the maps as published.

## J Conventions in the order parameters

Two choices in (12) deserve statement.

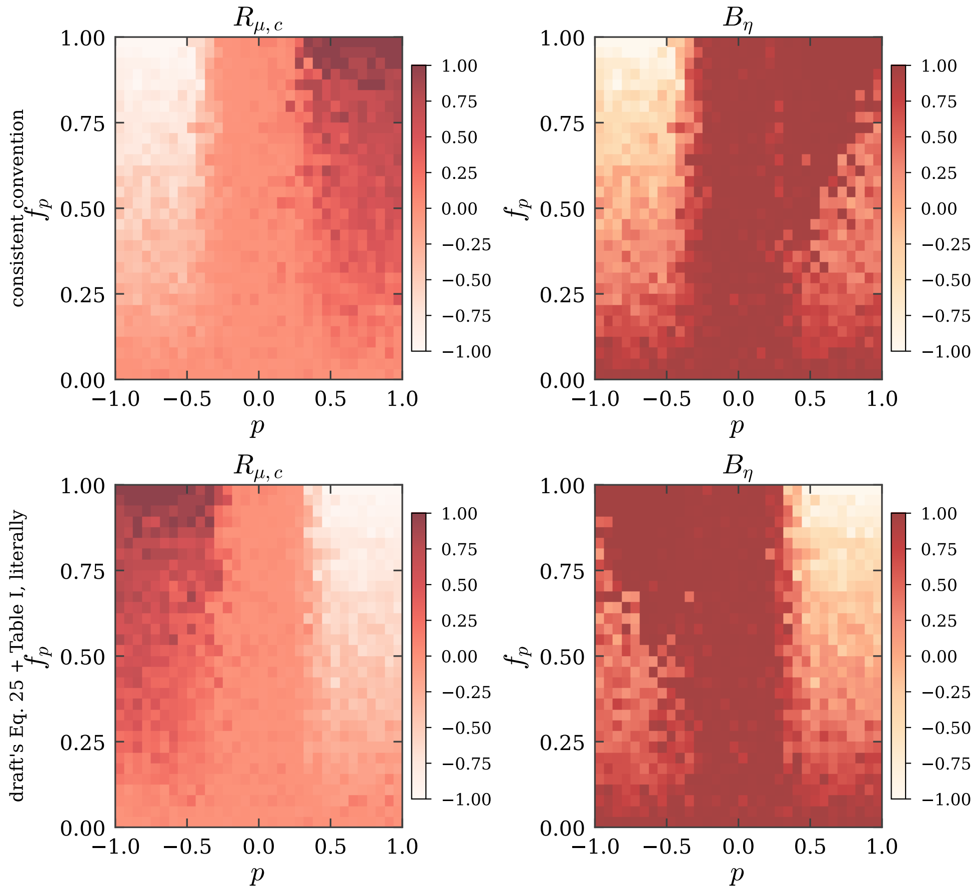



Figure 13: **The two readings of the prejudice field.** Trust–class correlation and trust balance over the $(p, f_p)$ plane under the convention used in this paper (top) and under the companion manuscript's table read literally with its Eq. 25 (bottom). The physics is identical; which half of the $p$ axis is discriminatory is not.

**The class indicator.** We use $G_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse discrimination it returns $-(N/2-1)/(N-1)$ rather than $-1$, and its range depends on $N$. Since the reverse-discrimination half of the phase diagram is a substantive part of the result, the signed indicator is the appropriate one. Both are implemented.

**Normalization.** All three averages in (10) and (12) run over the $N(N-1)/2$ unordered pairs, and the $\frac{1}{2}$ on the symmetrized trust puts all three in $[-1, 1]$. This matters because the three are read together as the colour channels of Figure 5; a literal reading of the model as previously written normalizes $R_{c,w}$ by $N(N-1)$ instead, capping it at $1/2$ and making the channels incommensurate. Both are implemented.

## K Parameters

The model needs six numbers. One is recovered from the companion manuscript, one is calibrated against its published figures, two are prior conventions, and two are free choices; all six are listed with their provenance rather than left implicit, since the manuscript states none of them. The accompanying code exposes each of them and its calibration script reproduces the comparisons below.

| symbol | meaning | value | provenance |
|---|---|---|---|
| $K$ | dimension of the issue embedding | 30 | recovered (see below) |
| $N$ | agents in the society | 40 | chosen; converged (see below) |
| $P$ | issues on the agenda | 5 / 100 | chosen: $\alpha = P/K$ either side of 1 |
| $\Delta t$ | interactions per ordered pair | 500 | calibrated (see below) |
| $C_0, V_0$ | initial uncertainties | $I$, 1 | uninformative prior |
| $\mu_0$ | initial distrust | $\mathcal{U}(-1, 1)$ | gives $B_\rho \approx B_\eta \approx 0$ at $t = 0$ |

**$K$ is recovered, not chosen.** The companion manuscript reports balance trajectories at $\alpha \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$. These are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$, which fixes $K = 30$.

**$\Delta t$ is calibrated against the published trajectories.** The dynamics anneals rather than reaching a stationary state, so the measurement time matters and the companion manuscript does not state it (its text carries a literal placeholder in place of the value). We scan $\Delta t$ and compare the endpoints of all nine balance trajectories against the published ones:

| $\Delta t$ (interactions per ordered pair) | 60 | 125 | 250 | **500** | 1000 |
|---|---|---|---|---|---|
| rms distance to published endpoints | 0.63 | 0.31 | 0.15 | **0.13** | 0.15 |

The minimum is flat-bottomed and sits at $\Delta t = 500$, which we use throughout. Seven of the nine endpoints then agree to within about 0.1, and $B_\rho$ matches the three largest agendas almost exactly, but no single $\Delta t$ reproduces all nine: two middle curves ($\alpha = 0.23$, $0.33$) sit above ours, and the two largest agendas have a lower published $B_\eta$ than we obtain at any $\Delta t$ that fits the rest. Either the society size differs from ours or those middle curves, which overlap within a few pixels in the printed figure, cannot be digitized reliably; we cannot distinguish the two. Dropping the ambiguous readings does not move the selected value. All the $\Delta t$-independent checks pass: the above/below-diagonal ordering with its crossover at $\alpha \approx 1.7$, the sharp sign flip of $R_{\mu,c}$ at $p \approx 0$, and the $R_{c,w}$ wedge confined to large $p$ and large $f_p$.

**$N = 40$ is large enough.** Repeating five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realizations each, the order parameters at $N = 60$ differ from those at $N = 40$ by at most 0.02 ($R_{\mu,c}$, $R_{c,w}$), 0.03 ($B_\rho$) and 0.05 ($B_\eta$), while $N = 20$ differs by up to 0.24. The maps are therefore converged in $N$ at $N = 40$, and the cost of going further is steep: total work grows as $N^3$.

**Numerical safeguards.** Three floors are applied, none of which the model prescribes: on the evidence $Z$, whose exact modulation functions diverge in the deep dissonance corners; on the trust variance $V$, which $F_V < 0$ drives monotonically towards zero; and on the covariance update, clipped to the boundary at which $C + a(C\hat{x})(C\hat{x})^\top$ ceases to be positive semi-definite. The last is the only one that can alter a trajectory, and it never triggered in the runs reported here.